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Q1. Reverse a list

Code

```
#include<stdio.h>

int main()
{
    int n,a[100],i;
    scanf("%d",&n);
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);
    for(i=n-1;i>=0;i--)
        printf("%d ",a[i]);
    return 0;
}
```

Input

5

1 2 3 4 5

Output

5 4 3 2 1

Q2. Reverse a string

Code

```
#include<stdio.h>

int main()
{
    int n,rev=0,rem,temp;
    scanf("%d",&n);
    temp=n;
    while(n>0)
    {
```

```

        rem=n%10;

        rev=rev*10+rem;n=n/10;
    }
    if(temp==rev)
        printf("Pallindrome");
    else
        printf("Not Pallindrome");
    return 0;
}

```

Input

121

Output

Pallindrome

Q3. Searching

Code

```

#include<stdio.h>

int main()
{
    int a[100],n,i,key;
    int low=0,high,mid;
    scanf("%d",&n);
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);
    scanf("%d",&key);
    high=n-1;
    while(low<=high)
    {
        mid=(low+high)/2;
        if(a[mid]==key)
        {
            printf("Element Found");

```

```

        return 0;
    }
    else if(key<a[mid])
        high=mid-1;
    else
        low=mid+1;
    }
    printf("Element Not Found");
    return 0;
}

```

Input

5

10 20 30 40 50

30

Output

Element Found

Q4. Searching

Code

```

#include<stdio.h>

int main()
{
    int a[100],n,i,key,count=0;

    scanf("%d",&n);
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);
    scanf("%d",&key);
    for(i=0;i<n;i++)
    {
        if(a[i]==key)
            count++;
    }
}

```

```
    printf("%d",count);  
    return 0;  
}
```

Input

5

2 4 2 6 2

2

Output

3

Q5. Array

Code

```
#include<stdio.h>  
  
int main()  
{  
    int a[100],n,key;  
    int low,high,mid;  
    scanf("%d",&n);  
    for(int i=0;i<n;i++)  
        scanf("%d",&a[i]);  
    scanf("%d",&key);  
    low=0;  
    high=n-1;  
    while(low<=high)  
    {  
        mid=(low+high)/2;  
        if(a[mid]==key)  
        {  
            printf("Found");  
            return 0;  
        }  
        else if(key<a[mid])
```

```

        high=mid-1;
    else
        low=mid+1;
}
printf("Not Found");
return 0;
}

```

Input

5

10 20 30 40 50

40

Output

Found

Q6. Merging

Code

```

#include<stdio.h>

int main()
{
    int a[50],b[50],c[100];
    int n1,n2,i,j;
    scanf("%d",&n1);
    for(i=0;i<n1;i++)
        scanf("%d",&a[i]);
    scanf("%d",&n2);
    for(i=0;i<n2;i++)
        scanf("%d",&b[i]);
    for(i=0;i<n1;i++)
        c[i]=a[i];
    for(j=0;j<n2;j++)
        c[i++]=b[j];
    for(i=0;i<n1+n2;i++)

```

```
        printf("%d ",c[i]);  
    return 0;  
}
```

Input

4

1 2 3 4

5

5 6 7 8 9

Output

1 2 3 4 5 6 7 8 9

Q7. Largest and Smallest element

Code

```
#include<stdio.h>  
  
int main()  
{  
    int a[100],n,i,max,min;  
    scanf("%d",&n);  
    for(i=0;i<n;i++)  
        scanf("%d",&a[i]);  
    max=min=a[0];  
    for(i=1;i<n;i++)  
    {  
        if(a[i]>max)  
            max=a[i];  
        if(a[i]<min)  
            min=a[i];  
    }  
    printf("Largest = %d\n",max);  
    printf("Smallest = %d",min);  
}
```

Input

5

10 25 6 40 22

Output

Largest = 40

Smallest = 6

Q8. Second Largest Element

Code

```
#include<stdio.h>

int main()
{
    int a[100],n,i;
    int largest,second;
    scanf("%d",&n);
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);
    largest=second=a[0];
    for(i=1;i<n;i++)
    {
        if(a[i]>largest)
        {
            second=largest;
            largest=a[i];
        }
        else if(a[i]>second&& a[i]!=largest)
        {
            second=a[i];
        }
    }
    printf("%d",second);
    return 0;
}
```

Input

5

10 40 20 30 50

Output

40

Q9. Occurrence of an elemnet

Code

```
#include<stdio.h>

int main()
{
    int a[100],n,i,key;
    int first=-1,last=-1;
    scanf("%d",&n);
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);
    scanf("%d",&key);
    for(i=0;i<n;i++)
    {
        if(a[i]==key)
        {
            if(first==-1)
                first=i;
            last=i;
        }
    }
    if(first==-1)
        printf("Not Found");
    else
    {
        printf("First= %d\n",first);
        printf("Last=%d",last);
    }
}
```



```
    }  
    return 0;  
}
```

Input

6

10 20 30 20 40 20

20

Output

First= 1

Last=5

Q10. Missing Element

Code

```
#include<stdio.h>  
  
int main()  
{  
    int a[100],n,i;  
    scanf("%d",&n);  
    for(i=0;i<n;i++)  
        scanf("%d",&a[i]);  
    for(i=0;i<n-1;i++)  
    {  
        if(a[i+1]!=a[i]+1)  
        {  
            printf("%d",a[i]+1);  
            break;  
        }  
    }  
    return 0;  
}
```

Input

5

1 2 4 5 6

Output

3